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Roll No

TX-5005(1)-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Fibre Science - II

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Describe the manufacturing process of carbon fibre from PAN.
b) What is Bi-component fibre? How bi-component fibres are manufactured?
2. a) Write down the structural changes in fibre on heating.
b) Define following:
 - i) Specific heat
 - ii) Melting
 - iii) Heat setting
3. a) Explain the various end uses of glass fibre.
b) Explain various physical properties of Spandex and Teflon fibres.
4. a) Explain the effect of impurities on electrical resistance of fibre.
b) Explain the different thermal parameters in detail.

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5. a) What is Glass fibre? State the properties and uses of glass fibre.
b) How glass fibre is manufactured? Explain in detail its manufacturing process.
6. a) Explain how static charge is generated. Also state what are the problems due to static charge generation.
b) Explain the various factors affecting the heat setting behaviours of textile material.
7. a) What is Air-jet texturing process? Explain it with suitable diagram.
b) Write down the Tg and melting points of Polyester, Nylon, Acrylic, Polypropylene and Polyethylene fibers.
8. a) Explain the manufacturing method of Spectra in detail.
b) What is first order and second order transition with their importance?

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